

PRACTICAL 10:

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ stat inode_test.txt
  File: inode_test.txt
  Size: 36             Blocks: 8          IO Block: 4096   regular file
Device: 8,48    Inode: 13737        Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/nirvighna)   Gid: ( 1000/nirvighna)
Access: 2026-09-29 18:53:53.334582538 +0000
Modify: 2026-09-29 18:53:53.338582539 +0000
Change: 2026-09-29 18:53:53.338582539 +0000
Birth:  2026-09-29 18:53:53.334582538 +0000

nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ln inode_test.txt hard_link.txt
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls -li inode_test.txt hard_link.txt
13737 -rw-r--r--  2 nirvighna nirvighna 36 Sep 29 18:53 hard_link.txt
13737 -rw-r--r--  2 nirvighna nirvighna 36 Sep 29 18:53 inode_test.txt
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ln -s inode_test.txt soft_link.txt
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls -li inode_test.txt hard_link.txt so
ft_link.txt
13737 -rw-r--r--  2 nirvighna nirvighna 36 Sep 29 18:53 hard_link.txt
13737 -rw-r--r--  2 nirvighna nirvighna 36 Sep 29 18:53 inode_test.txt
13738 lrwxrwxrwx  1 nirvighna nirvighna 14 Sep 29 18:54 soft_link.txt -> inod
e_test.txt
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls -li inode_test.txt hard_link.txt so
ft_link.txt
13737 -rw-r--r--  2 nirvighna nirvighna 36 Sep 29 18:53 hard_link.txt
13737 -rw-r--r--  2 nirvighna nirvighna 36 Sep 29 18:53 inode_test.txt
13738 lrwxrwxrwx  1 nirvighna nirvighna 14 Sep 29 18:54 soft_link.txt -> inod
e_test.txt

nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ readlink soft_link.txt
inode_test.txt
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ cat hard_link.txt
This is Session 10 inode practical.
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ cat soft_link.txt
This is Session 10 inode practical.
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ stat inode_test.txt
  File: inode_test.txt
  Size: 36             Blocks: 8          IO Block: 4096   regular file
Device: 8,48    Inode: 13737        Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/nirvighna)   Gid: ( 1000/nirvighna)
Access: 2026-09-29 18:59:50.118548043 +0000
Modify: 2026-09-29 18:53:53.338582539 +0000
Change: 2026-09-29 18:54:41.662587689 +0000
Birth:  2026-09-29 18:53:53.334582538 +0000
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ |
```

```
nirvighna@DESKTOP-VS57D2  ×  +  ▾
GNU nano 7.2 src/mmap_file.c *
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/mman.h>
#include <sys/stat.h>
#include <string.h>

int main()
{
    int fd;
    struct stat st;
    char *mapped;

    fd = open("mmap_test.txt", O_RDWR | O_CREAT, 0644);

    if (fd == -1)
    {
        perror("open");
        return 1;
    }

    if (ftruncate(fd, 100) == -1)
    {
        perror("ftruncate");
        close(fd);
        return 1;
    }

    if (fstat(fd, &st) == -1)
    {
        perror("fstat");
        close(fd);
        return 1;
    }

    mapped = mmap(NULL, st.st_size,
```

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^\ Replace	^U Paste	^J Justify

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ gcc -Wall -Wextra -g src/mmap_file.c -
o bin/mmap_file
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ./bin/mmap_file
Data written using mmap():
Hello from mmap! This file was accessed using memory mapping.
mmap operation completed successfully.
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ |
```

```
nirvighna@DESKTOP-VS57D2  ×  +  ▾
GNU nano 7.2 src/normal_io.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int fd;
    char buffer[101];
    ssize_t bytes_read;

    fd = open("mmap_test.txt", O_RDONLY);

    if (fd == -1)
    {
        perror("open");
        return 1;
    }

    bytes_read = read(fd, buffer, 100);

    if (bytes_read == -1)
    {
        perror("read");
        close(fd);
        return 1;
    }

    buffer[bytes_read] = '\0';

    printf("Data read using read():\n");
    printf("%s", buffer);

    close(fd);

    return 0;
}
```

Help **Write Out** **Where Is** **Cut** **Execute**
Exit **Read File** **Replace** **Paste** **Justify**

12:33 AM 9/30/2026

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ cat mmap_test.txt
Hello from mmap! This file was accessed using memory mapping.
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls -lh mmap_test.txt
-rw-r--r-- 1 nirvighna nirvighna 100 Sep 29 19:02 mmap_test.txt
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls -l bin/mmap_file
-rwxr-xr-x 1 nirvighna nirvighna 19168 Sep 29 19:02 bin/mmap_file
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ nano src/normal_io.c
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ gcc -Wall -Wextra -g src/normal_io.c -
o bin/normal_io
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ gcc -Wall -Wextra -g src/normal_io.c -
o bin/normal_io
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ cat hard_link.txt
This is Session 10 inode practical.
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ cat soft_link.txt
This is Session 10 inode practical.
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ |
```